

```
PS C:\Users\rajak\OneDrive\Desktop\docker practice> cd './terraform practice'
```

```
PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> terraform validate
```

Success! The configuration is valid.

```
PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> terraform validate
```

Success! The configuration is valid.

```
PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> terraform apply --auto-approve
```

aws_iam_user.demouser1: Refreshing state... [id=sjcedemo]
aws_instance.testec2: Refreshing state... [id=i-0aaea82e2831fab0d]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
-/+ destroy and then create replacement

Terraform will perform the following actions:

```
# aws_instance.testec2 must be replaced
```

```
-/+ resource "aws_instance" "testec2" {
```

~ arn	= "arn:aws:ec2:us-east-1:796498482092:instance/i-0aaea82e2831fab0d" -> (known after apply)
~ associate_public_ip_address	= true -> (known after apply)
~ availability_zone	= "us-east-1c" -> (known after apply)
~ disable_api_stop	= false -> (known after apply)
~ disable_api_termination	= false -> (known after apply)
~ ebs_optimized	= false -> (known after apply)

▼ terraform practice

> .terraform
≡ .terraform.lock.hcl
ec2.tf
iam.tf
provider.tf
terraform.tfstate
terraform.tfstate.backup

```
1 resource "aws_instance" "testec2" {  
2     ami           = "ami-0b6d9d3d33ba97d99"  
3     instance_type = "t3.micro"  
4     key_name      = "key1"  
5     vpc_security_group_ids = ["sg-05103ef2db53871fa"]  
6  
7     tags = {  
8         Name = "demoec2"  
9         name = "demoinstance"  
10        team = "sjce-devops"  
11    }
```

Problems Output Debug Console Terminal Ports

powershell - terraform practice + -

PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> terraform apply --auto-approve

```
- tags_all           = {} -> null  
- throughput         = 125 -> null  
- volume_id          = "vol-0a52cb5b21c7e078e" -> null  
- volume_size        = 8 -> null  
- volume_type        = "gp3" -> null  
# (1 unchanged attribute hidden)  
}
```

```
~ secondary_network_interface (known after apply)  
}
```

Plan: 1 to add, 0 to change, 1 to destroy.

aws_instance.testec2: Destroying... [id=i-0aaea82e2831fab0d]

aws_instance.testec2: Still destroying... [id=i-0aaea82e2831fab0d, 00m10s elapsed]

aws_instance.testec2: Still destroying... [id=i-0aaea82e2831fab0d, 00m20s elapsed]

aws_instance.testec2: Still destroying... [id=i-0aaea82e2831fab0d, 00m30s elapsed]

aws_instance.testec2: Destruction complete after 32s

aws_instance.testec2: Creating...

aws_instance.testec2: Still creating... [00m10s elapsed]

aws_instance.testec2: Creation complete after 16s [id=i-0b3d009442791c66c]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.

● PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> terraform apply --auto-approve

> Outline

> Timeline

```
PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> terraform apply --auto-approve
```

```
Apply complete! Resources: 1 added, 0 changed, 1 destroyed.
```

```
● PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> terraform apply --auto-approve
```

```
aws_iam_user.demouser1: Refreshing state... [id=sjcedemo]
```

```
aws_instance.testec2: Refreshing state... [id=i-0b3d009442791c66c]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

~ update in-place

Terraform will perform the following actions:

```
# aws_instance.testec2 will be updated in-place
```

```
~ resource "aws_instance" "testec2" {
```

```
    id = "i-0b3d009442791c66c"
```

```
    ~ tags = {
```

```
      + "Name" = "demoec2"
```

```
      "name" = "demoinstance"
```

```
      "team" = "sjce-devops"
```

```
    }
```

```
    ~ tags_all = {
```

```
      + "Name" = "demoec2"
```

```
      # (2 unchanged elements hidden)
```

```
    }
```

```
    # (39 unchanged attributes hidden)
```


Explorer

docker practice

terraform practice

.terraform

.terraform.lock.hcl

ec2.tf

iam.tf

provider.tf

terraform.tfstate

terraform.tfstate.backup

provider.tf

iam.tf

ec2.tf X

terraform practice > ec2.tf > resource "aws_instance" "testec2" > [] vpc_security_group_ids > abc 0

1 resource "aws_instance" "testec2" {

2 ami = "ami-0b6d9d3d33ba97d99"

3 instance_type = "t3.micro"

4 key_name = "key1"

5 vpc_security_group_ids = ["sg-05103ef2db53871fa"]

6

7 tags = {

8 Name = "demoec2"

9 name = "demoinstance"

10 team = "sjce-devops"

11 }

Problems

Output

Debug Console

Terminal

Ports

powershell - terraform practice + v |

PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> terraform apply --auto-approve

}

Plan: 1 to add, 0 to change, 1 to destroy.

aws_instance.testec2: Destroying... [id=i-0b3d009442791c66c]

aws_instance.testec2: Still destroying... [id=i-0b3d009442791c66c, 00m10s elapsed]

aws_instance.testec2: Still destroying... [id=i-0b3d009442791c66c, 00m20s elapsed]

aws_instance.testec2: Still destroying... [id=i-0b3d009442791c66c, 00m30s elapsed]

aws_instance.testec2: Destruction complete after 32s

aws_instance.testec2: Creating...

aws_instance.testec2: Still creating... [00m10s elapsed]

aws_instance.testec2: Creation complete after 17s [id=i-0c97725fa129bf908]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.

PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> terraform apply --auto-approve

aws_iam_user.demouser1: Refreshing state... [id=sjcedemo]

aws_instance.testec2: Refreshing state... [id=i-0c97725fa129bf908]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.

PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice>

> Outline

> Timeline

Ln 5, Col 50

Spaces: 1

UTF-8

CRLF

EC2

>

Instances

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Successfully initiated starting of i-0aaea82e2831fab0d

Instances (1/2)

Info

Last updated 26 minutes ago

Refresh

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Filter Instance by attribute or tag (case-sensitive)

< 1 >

Settings

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
☒	demoec2	i-0b3d009442791c66c	Running	t3.micro	3/3 checks passed	us-east-1c	ec2-34-233-133
☐		i-0aaea82e2831fab0d	Terminated	t3.micro	-	us-east-1c	-



terraform practice > vpc.tf > ...

```

81 resource "aws_vpc_security_group_ingress_rule" "ssh" {
82     security_group_id = aws_security_group.allow_tls.id
83     cidr_ipv4         = "0.0.0.0/0"
84     from_port         = 22
85     ip_protocol        = "tcp"
86     to_port            = 22
87 }
88 resource "aws_vpc_security_group_egress_rule" "allow_all" {
89     security_group_id = aws_security_group.allow_tls.id
90     cidr_ipv4         = "0.0.0.0/0"
91     ip_protocol        = "-1"
92 }
93

```

Problems Output Debug Console **Terminal** Ports

powershell - terraform practice

```
PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice> terraform apply --auto-approve
aws_instance.testec2: Creating...
aws_vpc_security_group_ingress_rule.https: Creating...
aws_vpc_security_group_ingress_rule.ssh: Creating...
aws_vpc_security_group_ingress_rule.http: Creating...
aws_vpc_security_group_ingress_rule.https: Creation complete after 2s [id=sgr-0795ff1f3a41afdb0]
aws_vpc_security_group_ingress_rule.http: Creation complete after 2s [id=sgr-0dbe6c519496ee7cc]
aws_vpc_security_group_ingress_rule.ssh: Creation complete after 2s [id=sgr-0d557c16691c43419]
aws_instance.testec2: Still creating... [00m10s elapsed]
aws_instance.testec2: Creation complete after 18s [id=i-0b9c9d70924068f37]
```

Apply complete! Resources: 4 added, 0 changed, 0 destroyed.

```
PS C:\Users\rajak\OneDrive\Desktop\docker practice\terraform practice>
```

Timeline

EC2

- Dashboard
- AWS Global View
- Events
- Instances
- Instances
- Instance Types
- Launch Templates

Instances (1/1) Info

Last updated
2 minutes ago



Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set



Filter instance by attribute or tag (case-sensitive)



1



Name



Instance ID

Instance state



Instance type



Status check

Availability Zone



Public IPv4 DNS



demoec2

i-0b9c9d70924068f37

Running



t3.micro

Initializing

us-east-1c

-

VPC dashboard <

Filter by VPC ▼

[AWS Global View](#) ↗

▼ Virtual private cloud

[Your VPCs](#)

[Subnets](#)

[Route tables](#)

Your VPCs (1/1) [Info](#)

Last updated
1 minute ago



Actions ▼

Create VPC

Find VPCs by attribute or tag

Name = demo-vpc



Clear filters



1



Name



VPC ID



State



Encryption c... ▼

Encryption control ... ▼

Block Public... ▼

IP



demo-vpc

[vpc-0dba337ae20d4a630](#)

✓ Available

–

–

⊖ Off

10

Usage of /:	30.5% of 6.61GB	Processes:	118
Memory usage:	26%	Users logged in:	0
Swap usage:	0%	IPv4 address for ens5:	10.0.1.179

Expanded Security Maintenance for Applications is not enabled.

updates can be applied immediately.

enable ESM Apps to receive additional future security updates.
see <https://ubuntu.com/esm> or run: `sudo pro status`

The list of available updates is more than a week old.
to check for new updates run: `sudo apt update`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

```
buntu@ip-10-0-1-179:~$ date
```

```
Mon Oct  5 14:10:00 UTC 2026
```

```
buntu@ip-10-0-1-179:~$ whoami
```

```
buntu
```

```
buntu@ip-10-0-1-179:~$ cal
```

```
Command 'cal' not found, but can be installed with:
```

```
sudo apt install ncal
```

```
buntu@ip-10-0-1-179:~$ ls
```

```
buntu@ip-10-0-1-179:~$ █
```